

Institute of Actuaries of India

Subject SA3 – General Insurance

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) a) Key areas to be investigated are [1.5]
- Actual vs expected business volumes / premiums (gross and net) / exposure / mix
 - Actual vs expected claims experience – severity / frequency / mix / reporting and settlement delays
 - Actual vs expected expenses – policy expenses: fixed and variable/ business expenses and overheads
 - Actual vs expected investment income – impact on assets and liabilities

Divergence of actual claims experience from that expected in pricing basis is most likely to give rise to maximum potential benefit

[0.5]

[2]

b)

Key factor:

- Ascertain the time period to be considered to collect the data for investigation. In this case data during the lock down period between April and June is most likely to be considered. This would again depend on the geographical exposure as different states had different stages and periods of lockdown.
- Ascertain the time horizon over which the company intends to return any accrued benefit to the policyholders. This depends on the management view of future and its strategy whether short/medium or long-term

[1]

Assessments areas:

Premiums

[1]

- Potential reduction in new and renewal business – negative impact on revenues
- Fixed RI XL costs for ensuing year on business projections but lower than budgeted business volumes – negative impact on cash position?

Claims

[5]

- Motor OD:
 - Potential benefit because of lower accident rates as vehicle not plying on roads
 - But, should an event arise, severity could be higher due to longer periods in garage (particularly if daily cash benefits are provided), affected supply chains and non-availability of / expensive spares and labour
 - However possibly not much change in incidence from perils like flood, fire, theft
 - Higher than normal reporting lag expected due to lockdown and late discovery of damage due to fire perils / theft. However, if the assessment is being carried out in say September, most claims would have been reported.
 - Since OD is a short tailed line, the impact of benefit would be more apparent in a month or two after the large scale lockdown is over
- Motor TP:
 - Reporting delays are potentially further exacerbated on account of courts not

functioning

- Since there are restrictions on number of travellers in a vehicle, average number of victims per event could reduce
- Claim may not be payable if the vehicle were plying on the roads without a valid e-pass required at that point in time
- Claims severity could be higher if the mix of injury and death cases were different due to greater vehicle speeds. At the same time, the road accident victims during this period could be having lower income levels. Assessing impact on Motor TP claims severity could be a challenge
- Information such as the following could be used as proxies to assess the reduction in risk exposure, subject to necessary adjustments
 - OD claims experience in the lockdown compared to earlier period
 - Traffic density from Google Maps or any vendor who can provide such information in the lockdown compared to earlier period
 - Fuel sales (petrol, diesel etc.) in the lockdown compared to earlier period
 - Road accident statistics from government agencies during the lockdown compared to earlier period
- Property:
 - No change in risk exposure for residential property (householders' package). However, burglary risk would have expectedly reduced.
 - Business interruption due to pandemic was not covered under the property products. Could claims get more litigious in this situation increasing defense costs?
 - Fire claims propensity for small commercial risks could have reduced if there was no production or lower than usual utilization of capacity. Stocks would still be at risk for perils such as Fire, inundation, etc.
 - Unattended stocks could give rise to claims with a higher loss severity - e.g. rodent / termite attack; total loss of perishables; explosions with hazardous chemicals in stock
 - Strained economic activity could give rise to fraudulent claims from businesses

Expenses

[2]

- Potential savings in actual commissions doled out as mix of new business/renewals would have been most likely online/ direct as against via an agent/intermediary
- Potential reduction in per policy fixed costs like policy documentation and courier charges due to likely online delivery of the same
- There could also be a reduction in recurring overheads for the months in lockdown like electricity charges, printing and stationery due to work from home policies and branches/offices being shut. If the company had witnessed a higher employee attrition rate during the period, this could mean further savings in this area as well.
- However, overall reduction in business volumes due to lockdown could mean on an average a higher per policy fixed costs from other operational and business overheads like advertising, employee remuneration & benefits, rents, etc.
- Whether there are sliding scale ceding commissions – this could be higher due to favourable loss ratios
- A combination of the above may or may not mean an overall net potential benefit

Investment income

[2]

- Investment income and asset gains would have reduced due to
 - Reduced cash flow from premiums due to business slowdown
 - Poor asset performance due to stock market volatility witnessed during the period
 - Overall reduction in interest rates
- Potential benefit in the form of Motor TP case reserves accumulating at a lower interest rate coupled with lower inflation rates. Besides with higher unemployment incidence, the wage loss could also translate to lower reserves to be set aside on expected court awards.

[10]

c)

Insurance is about specified coverage during a specified period for a specified price

Price related benefits – premium refunds, premium reduction for same coverage and same period of cover

- Usually passed on at policy issuance or anytime during the policy period
- Premium can be refunded or placed in policyholder's credit to be offset with renewal premium. The latter could likely increase retention ratios
- Renewal premium discounts could also increase retention ratio

Coverage related benefits – increase in sum insured or reduction in quantum of excess/deductible for the same premium and same period of cover. This may be viewed as a way of deferring the sharing of benefit by the insurer i.e. a policyholder 'benefits' in the event of a claim, not otherwise. May be viewed as regressive, as claimants tend to benefit and non-claimants do not.

Period related benefits – extended period of cover for same premium and same coverage

- Automatic extension of policy period by a certain number of days to reflect the favourable experience during lockdown period

[3]

d)

Considerations:

- Principle of insurance
 - It may be argued that when unexpected adverse events do not entitle an insurance company to collect additional premiums from the policyholder, should favourable circumstances entitle the policyholder of premium refunds?
 - That said, just as any adverse experience could lead to corrective premium increases, sharing benefits of a favourable experience through premium discounts would seem only logical
- Gain competitive advantage
 - The insurer would be eyeing to gain first mover advantage by announcing such ahead of others
 - This can potentially increase retention and new business due to goodwill and positive

market reputation

- Business ethics and industry camaraderie
 - The insurer may want to first discuss in industry forums for transparency
 - Particularly since it is a systemic benefit accruing to all insurers not just this individual insurer
 - Greater onus for such transparency for larger insurers so that such a decision does not become onerous on other small/new insurers?
- Regulatory developments
 - The insurer may want to wait and watch for any regulatory developments
- Regulatory approvals
 - An important consideration for the insurer is that any changes in premiums, benefits or period of coverage would need prior regulatory approvals.
 - The likely turn-around-time involved to implementation would be a key consideration
- View of the management
 - Business scenarios in the short term could be different to medium to long term view.
 - Long term view could potentially offset any accrued benefits in the short term which can influence insurer's decision, for e.g., claim spikes in subsequent periods or continued low business volumes over a longer period than anticipated
- Policyholder expectations
 - Creating a precedent of premium refunds by a few insurers may prove to be too onerous for other insurers. Could this lead to increased complaints/grievances from policyholders of those insurers that do not choose to pass on the benefit?
- Moral hazard
 - Can announcement of decision for premium refunds alter policyholder behaviour? For e.g. withholding claim reporting till premiums are refunded and then reporting a claim
 - Could a premium discount on renewal disincentivise NCB protection claimant behaviour, even partially? Could this give rise to more small claims being reported than otherwise?
 - Of course, the extent of alteration in insured behaviour would depend on the quantum of refund / discount anticipated.
- Administrative costs
 - Administration and processing costs could outweigh the quantum of benefits to be passed on
 - The costs could be particularly significant if the benefits are not standard across policyholders, i.e. if the benefit varies based on:
 - Whether the policyholder has claimed during the policy period or not
 - Geographical location of the insured (considering lockdowns were state driven)
 - Policy being in-force during a period of reckoning or as on date of reckoning
 - Whether policyholder is given options to choose the form of benefit

[9]

e)

Altered business and risk profile (maximum 7 marks):

- Economic and social activity may remain subdued in near to medium term future compared to normal times. Therefore, in this lull period, market participants like individuals, service operators, transport lobbies, may be expecting or urging for lower Motor OD rates.
- Motor insurance accounts for over one-third of total premiums in non-life insurance

industry. Therefore, insurers cannot afford to lose its most vital volume base thus necessitating premium reduction to attract and retain customers.

- New automobile sales had seen a lull in 2019-20 that continued in 2020-21 due to the pandemic outbreak. New vehicle insurance that usually are a considerable portion of total premiums have been adversely affected. Besides, to tap into the existing vehicle market, companies need aggressive pitching by sales force and its agents on the field that have also been adversely affected due to the lockdown. Thus, the key strategy would likely be maximizing retention that would mean offering best/reduced rates to its existing customers to ensure they renew their policies.
- With most people staying indoors and use of vehicles drastically reduced, policyholders could consider renewing only mandatory TP cover and not purchase the OD cover. A premium reduction could incentivize such policyholders to also purchase OD cover.
- Competitors may also be considering reducing OD premiums to gain first mover advantage. Any insurer that acts late may pose itself risk of losing goodwill amongst its customers besides also losing existing and potential new business.
- Slowly normalcy is approaching. But some industry employees are continuing with work-from-home policy, in full or partial capacity. This could mean number of vehicles plying on the road are expected to be lower in the 'new normal'. This altered risk exposure would mean a need to review existing prices.
- There could be regulatory advice for all insurers to revisit their pricing models in Motor OD
- A few companies have launched usage-based motor insurance on pilot basis. So, for these policies the new circumstances by design could mean potential reduction in premiums. However, the motor insurance market, by and large, is yet made up of regular comprehensive, liability only, standalone OD products.
- More use of own vehicles as people would be anxious of using public transport or group vehicles (e.g. school/office commuting buses), this potentially means increased exposure and hence more claims
- The above applies not only for intra-city but inter-city travel as well – greater highway traffic could also mean higher claims of potentially greater severity due to higher speeds
- Vehicles have been used little for months – more maintenance issues leading to greater accidents?
- Drivers that have hardly driven for months because of lockdown or those that drove less often now compelled to drive more often due to virus transmission concerns in public transport could have become rusty? This could mean poorer judgment and hence more accidents?
- Even if more and more choose to use aggregator cab services, the exposure would still increase albeit in taxi segment instead of in private vehicle segment
- Claims costs could be higher due to non-availability / costlier labour, affected supply chains leading to spare parts getting dearer, more time vehicle spends in a garage could mean higher claims particularly where 'daily cash' benefit is allowed (work-from-home may not make the claim inadmissible!)
- Sluggish automobile sales, poorer renewal ratios could affect volumes considerable thus affecting company's liquid cash position.
- This may necessitate a higher ticket size per policy sold
- If number of policies is lower than before, expense loading per policy might go up
- Lower interest rates may lead to lower cost of capital as also lower investment income
- Reinsurance cost might increase or decrease, it needs to be considered accordingly,

ceding commissions may be lower

- A combination of above i.e. reduced revenues, expenses not reducing proportionately, lower interest rates and erosion of asset value, could erode shareholder value.

Any revision in premiums or rating models would require prior regulatory approval

[1]

[8]

ii) a)

Risk Margin: No explicit requirement except for “arrive at an estimate that s/he believes is adequate to meet the future liabilities”

Discounting: “The estimation process shall not discount the estimated future development of paid claims to the current date”.

[2]

b)

The following observation is correct: “The amount paid during settlement in future could be greater than the provision held today since more interest will be accrued”.

However, future interest accrual on outstanding claims are provided in the form of IBNER. The additional interest which will be accrued later and paid out in the end, that component is provided under “IBNER”.

Hence, the overall provisions for unpaid claims are expected to be adequate.

The provisions are NOT discounted w.r.t reported but outstanding claims.

Under IFRS 17, discounting will be performed towards those total claim payment projections for all unpaid claims, whether they are reported or not.

[3]

c)

Under “General Measurement Model”:

- one of the individual blocks is the “future cashflows that arise from provision of insurance”. This covers both expired and unexpired risks.
- Further, for unexpired risks, IFRS17 requires CSM. In other words, it is the expected profit from unexpired risks. It is released as profit over time as coverage period elapses.
- If there is any premium deficiency, it means that the UPR is inadequate to cover sum of expected claim costs, related expenses and maintenance costs (related to claims handling) exceeds related reserve for unexpired risks.
- In other words, CSM is negative. IFRS17 requires such contracts to be identified as “Onerous contracts”. In such cases, CSM is set to zero, effectively recognizing the loss immediately. PDR also addresses the same purpose.

Under “Premium Allocation Approach”:

- Liability for unexpired risks is the premiums unearned net of insurance acquisition costs. This is similar to UPR.
- This does not apply to a group of insurance contracts which are onerous.

[4]

d)

Current Regulatory basis:

- Unpaid claim estimates will reduce since the interest component of claim payments will reduce due to lower interest rates.
- UPR is not impacted by interest rates.

IFRS 17:

- Unpaid claim estimates on undiscounted basis will reduce since the interest component of claim payments will reduce due to lower interest rates.
- Since discounting will also be done with similar rates, the reduction in rupee terms will be lower on account of lower rates.
- The above two will offset each other, assuming that there are no big fluctuations in interest rates.
- As far as liability for remaining coverage is concerned, there won't be any impact of interest rates if Premium Allocation approach is used.
- With regards to 'Risk Adjustment', IFRS 17 does not prescribe unlike Solvency II. In Solvency II, the liabilities are loaded with a risk margin which is calculated using a prescribed cost of capital rate of 6%. If risk adjustment under IFRS 17 is linked to cost of capital, the reduction in interest rates could translate to a reduction in cost of capital thereby reducing the risk adjustment.

Liabilities are Inflation Sensitive:

- In the context of Motor Third Party, amounts awarded by court are linked to income of the victim at the time of the event and prospects for wage growth. Income levels are correlated with overall inflation. If medium to long term future inflation outlook is low, prospects for wage growth may be revised downwards in future.
- Lower interest rates may also be associated with lower inflation rates. Since liabilities are inflation sensitive, overall increase in award amount is likely to be low.

However, Award amounts may not be impacted by lower future inflation on liabilities pertaining to expired risks. However, if there is any future judgment which considers lower prospects for wage growth, then it might impact.

[7]

[48 Marks]

Solution 2:

- i) *Key factors for Indian general insurance company (Regulatory considerations - maximum 4 marks):*

Meeting the minimum regulatory requirements...

...of having an available solvency margin (ASM) at least 1.5 times the required solvency margin

(RSM)

...breach of this control level of solvency will attract regulatory intervention

RSM will be dependent on the new company's business plan

...LoBs expected to be written, premium volumes

...business mix – certain classes (such as motor, liability and health) attract a higher charge relative to others (such as fire and engineering)

...target loss ratios for the classes written

...extent of outwards reinsurance used – cost and loss recoveries

ASM is calculated as the excess of admissible assets over liabilities, aggregated across policyholder and shareholder funds. This will broadly reflect the total capital infused to date, augmented (or eroded) by the accumulated profits (or loss), less fixed assets net of depreciation and other non-admissible assets per regulatory norms.

Meeting just the bare minimum 1.5 times is risky should actual experience deviate from the drafted business plan e.g. loss experience worsens or business mix tilts towards riskier classes. Also one would be on the regulator's watchlist.

IRDAI regulations on start-up capital – minimum paid up capital is currently INR 100 Cr for non-life insurance

Statutory solvency capital requirements on an ongoing basis– Currently minimum solvency ratio to be maintained is 1.5 and is monitored through quarterly reporting.

Any other alternate regulatory capital assessment criteria to be met – In India Economic capital assessment is also a mandatory requirement to assess capital sufficiency on an economic basis
[maximum 4 marks]

Other considerations:

- Internal board approved capital policy and internal minimum criteria to be met over and above regulatory minimum
- Risk appetite and Board risk tolerance statement on capital adequacy and solvency ratio at any given point in time, sometimes with a metric to avoid potential over capitalisation (simple e.g. solvency ratio not to fall below say 1.7 and not exceed say 2.5 at a desired level of probability)
- Underwriting strategy, expected business composition - concentration and accumulation, and expected exposure to large risks and high-risk category businesses
- New business strain and scale of start-up costs and expected year of break even as per business plan
- Planned capital infusions as per the insurer's business plan in the short to medium term could ease immediate capital infusion
- Distribution channels e.g. significant use of intermediaries / brokers could mean locking up of cash for months, necessitating larger initial capital
- Exposure to certain large scale Government insurance schemes or significant coinsurance business (as follower) could also mean locked up premiums and hence

need for greater initial capital

- Additional capital can provide greater flexibility with investment policy, reinsurance purchase and business strategy; Investment policy objectives and desired asset exposure
- Availability of additional reinsurance to help reduce capital requirements and trade-off between cost of reinsurance and cost of raising additional capital
- Competition benchmark for similar/ comparable business and risk exposure
- Rating agencies are likely to award a higher rating to well capitalised companies with a robust solvency ratio. What rating level is desired by management?
- Potential policyholders may not wish to purchase policies if the company's rating is poor due to low capital levels. This may adversely impact premium volumes.
- Expected cost of raising capital in the foreseeable future vis-à-vis current
- Opportunity cost of capital employment in insurance business vis-à-vis other opportunities, in this case say scaling up its existing e-commerce business further

[maximum 4 marks]

[max 4 marks for regulation related points; 0.5 marks for all points suitably explained]

[8]

ii)

Investment objective:

[1.5]

Main objective is to maximise investment return whilst meeting all contractual obligations and statutory norms, alongside addressing the uncertainties involved by keeping in consideration:

- the volume, type and classes of business planned to be written by New Age Insurance Co
- choice of assets to match with characteristics of liabilities

Internal:

Nature of liabilities

[1.25]

- Nature of liabilities – real / inflation linked or fixed in money terms
- Term of liabilities – short – tailed or long – tailed
- Level of uncertainty in amount and timing of liabilities
- Potential liabilities in a foreign currency – e.g. travel or export credit?
- Risk appetite and underwriting strategy of insurer

Investments and reserves

[1.25]

- Assets – their liquidity and marketability and how well they match with underlying liabilities to meet claim payments & expenses as and when they arise
- Volatility of asset values and investment returns relative to risk appetite
- Diversification of investment portfolio to manage risk
- Not all funds will be immediately available for investment. Need to allow for balances held by third parties and reinsurance arrangements
- Size of free reserves and level of investment freedom to optimise investment returns
- Shareholders' & investors' expected return on investment
- Investment expertise available in-house

External:

Stakeholder requirements (Regulator/Accounting & Audit/credit rating agency) [1]

- admissibility of certain assets for solvency calculations
- prescribed minimum and/or maximum percentage of investment assets to be held in certain types of investment
- Accounting standards and accounting valuation basis of investments
- Asset segmentation criteria of rating agencies that impact credit rating

Market norms and outlook [0.5]

- Industry norms and investment strategy of competitors in the Indian GI space
- Any legislative developments expected
- Future economic outlook on performance of different asset categories

Other [0.5]

- Tax treatment and net returns from different types of investment
- Asset availability in the market

[Maximum 5 marks] [5]

iii)

Future returns from the stock market may be lower than those expected by the CFO

Equities are less secure – higher risk of default especially with the current low level of the stock market

Equities experience greater volatility of returns, so a risky investment type

Equities better matched for longer term liabilities. A concern if most business planned to be underwritten is short tailed.

Equities may not be very liquid to fund general insurance claim payments that are typically met at a short notice. Realising a large amount of stock at a short notice may even yield lesser than expected realisation proceeds.

Regulatory restrictions on maximum percentage of investments that may be held as equities and consideration of value admissible for solvency [3]

iv)

Key benefits of technical pricing:

- Technical pricing based on an actuarial rating model would help capture multiple features that an underwriter may not be able to possibly account for at every quote and thus would more accurately reflect the risk. Besides premiums purely based on underwriter's judgment would be subjective whereas that based on a rating model would be more objective and trackable.
- A rating model can more effectively incorporate any rate adjustments for changes in assumptions like:

[1]

...expenses, reinsurance cost, investment income, cost of capital
 ...market trends as changes in claim propensity, claim cost and expense
 inflation, judicial inflation
 ...loadings for large losses, catastrophes and latent claims
 ...observed trends in attritional claim frequency
 ...changes in exposure, limits written, deductible/excess levels
 ...variation in coverage or policy terms and conditions
 ...any changes in regulation that may impact pricing
 ...allowing for cross-subsidies on the account

[3]

- A technical rating model can be seamlessly digitized with minimum human intervention. Particularly where products are made available online or through web aggregators, relying on underwriter inputs for each quote is not only cumbersome but impractical too.

[1]

- With a rating model, various rating factors can be captured along with their interactions, if any, to arrive at the final premium. These details would help in data mining for future analyses, e.g. business segmentation, exposure mix (concentrations and aggregations), risk characterization and categorization

[1]

- Regulations require a technical actuarial pricing note signed off by the actuary

[1]

- Other benefits (0.5 marks each):

- best suited for retail products like motor (OD), health etc. as volumes (no. of policies) are expected to be too big to be able to be individually quoted by the underwriter
- free up senior underwriters' time to concentrate on large/complex/non-standard risks where an underwriters' experience and judgment can be best utilized
- facilitate tracking of rating factors and corresponding rates on an account / allows for back-testing and review of the model to assess its appropriateness as experience emerges
- bring consistency in pricing / less subjective fluctuations arising from change in underwriter or change in underwriter's views with time
- instill greater confidence in reinsurers and may help reduce the cost of reinsurance and/or obtain greater cover
- better address regulatory concerns around underwriting standards and rating mechanism
- manage the risk of concentration of knowledge and complete reliance on the underwriter for pricing

[maximum 6 marks]

[6]

- v) A periodic rate monitoring and reporting mechanism is a useful tool in business analysis and can bring forth a host of benefits to the insurer:

A formal rate monitoring report will ideally capture...

...rate changes, actually achieved versus benchmark

...premium volume written versus plan

...new business volume

...renewal volume and retention rates versus target

[1]

...lost business

Additionally, it could capture the bound-to-technical ratio of the business underwritten, which will reflect the extent to which underwriters deviate from the actuarial price and will help monitor underwriter performance [0.5]

These metrics could be captured....

...over time e.g. month-on-month, to highlight movements and review the impact of any corrective actions taken

...by LoBs, product type, geographical spread etc. to identify segments doing well and those that are not

...comparison to business plan targets set for premium volumes, rate change etc. will identify early on the need and extent of any strategic changes [1.5]

This information will not only provide feedback to underwriters for improvements to the pricing process, but also help with actuarial reserving, capital modelling and business planning. It will provide directional guidance to management and help them with strategic decision making. [0.5]

It will provide information to understand accounts better and help with portfolio optimisation by...

...focusing on renewals of large profitable accounts

...identifying business concentration and accumulation of risks in certain areas

...defining threshold targets for underwriters to achieve at least x% rate change upon renewal for the segments identified as poorly performing

...identifying particularly bad risks to be avoided in future [1.5]

Formally documented rate monitoring process is also likely to be favourably viewed by rating agencies, regulators, brokers, reinsurers and shareholders. This may help negotiate better rates and/or terms with these external stakeholders. [0.5]

A well-documented process provides a clear audit trail allowing for better internal controls and smooth handover in the event of change to underwriting staff [0.5]

[maximum 4 marks]

[4]

vi)

Prefix of '+' or '-' or '=' indicate likely favourable or unfavourable or neutral impact of the new strategy

Any other rational points to be awarded marks

- Premiums (Financial projections and Business volumes):
 - + Generation of new business could be steady with a readily accessible client base; however, it would significantly depend on:
 - ...website design and popularity
 - ...advertising campaign success
 - ...internet usage and penetration across different cities/towns in the country
 - + Retention business is expected to be better in the new model due to ease of renewals leading to...
 - ...reduced lapses and hence more stable premium income in subsequent years

- ...reduce need for marketing and follow-ups over long term and hence reduced expenses
 - - Solely relying on ecommerce customer base may lead to reduced overall volume (no. of policies) and premiums – market outside the captive base would no longer be accessible
 - - Premiums can show an increasing trend in short to medium term as the captive base gets tapped but could gradually reach saturation as the ecommerce base reaches an equilibrium state
 - Other unfavourable impact can be:
 - - Coinsurance business that is usually through brokers would not be accessible in new model thus affecting business volume
 - - Need to draw a plan to ensure rural and social sector obligations would be consistently met in new model
- Financial projections: Claims /Loss Experience
 - = Claim frequency, severity and overall loss cost for this targeted customer segment likely to be different from industry experience.
 - = If New Age co. expects relatively better experience, it would either result in lower premium rates or higher underwriting profit for the same premiums, and vice-versa.
 - - This is difficult to assess at the outset without any historical experience and premium rates will have to be quickly adjusted as loss experience emerges
- Financial projections: Variable expenses
 - + Savings on commissions and reward payable to agents, brokers and other insurance intermediaries.
 - = However, if insurer intends to utilize online web aggregator platforms then commission payments may remain similar
- Financial projections: Fixed expenses and overheads
 - - Fixed expenses in the form of IT costs would be higher for...
 - ...system development and maintenance
 - ...IT equipment
 - ...expert staffing, recruitment
 - ...back-up and recovery
 - ...training
 - ...digital safety and cyber risk management
 - = tapping only the e-commerce clientele as suggested by the CEO, may not need aggressive advertising however if they desire to increase online traffic from clients beyond the in-house client base then
 - greater advertising costs would need to be incurred to increase visibility and for brand building
 - wide cross-section of advertising media e.g. in newspapers, magazines, billboards, television, search engines, price comparison websites etc. would need to be continually incurred in the formative years to promote the new company
 - + These expenses should be higher than existing budgeted in the initial years but gradually should fall in the long term as the company is established and well known, and targeted volumes are achieved
 - + The cost structure would over time likely to shift to more fixed expenses and

less variable expenses thus potentially reducing the expense ratio as volumes expand

- Financial projections: Per policy/claim expense
 - + Claim handling, loss adjustment and policy admin expenses may reduce...
...due to the increased use of online proposal processing, issuance of e-policies, online collection of premiums
...potentially managing smaller claims digitally over emails and settlement of claims through online payment modes
...bulk transactions could mean reduced payment gateway fees
 - - More susceptible to online fraud (phishing) and cyber risks (online uploads/data privacy/payment gateway security) hence higher per policy expense in building firewalls/cyber safety measures
 - = Reduction in overall expenses per policy will be contingent on sufficient business volumes being generated through online sales alone
- Financial projections: Reinsurance
 - - Should reinsurers perceive greater risk due to this business strategy on account of fraud, concentration of risk, deviation from industry norms, or worse loss experience then cost of reinsurance cover is likely to go up
- Financial projections: Investment Income
 - There is potential to earn more investment income relative to competitors as no premiums are locked away with brokers and intermediaries. The benefit could either translate to lower premium rates or higher profit margin.
- Line of Business Mix (Business volumes and financial projections)
 - = The online business model would be suited for class rated / retail product lines like Motor, Health, PA, Travel ...
 - = This likely concentration of business volumes in retail lines could significantly impact the projected financials (premiums/claims/expenses/capital projections and so on)
- Pricing impact (Business volumes)
 - - Premiums would need to be more competitive (lower) as online buyers are usually more 'discerning', and therefore they are likely to compare rates and coverages available across the market before purchase or renewal.
 - - Reduced average ticket size in the new model would mean lower overall revenue
- Target segment and policyholder characteristics (Business volumes & loss experience)
 - = Insured mix would be different than that of the wider overall market (likely to be seen in a typical sourcing model), i.e.
...possibly younger age groups
...different geographical spread (areas with higher internet penetration)
...skewed to urban population than rural customer base
...likely more educated, informed and tech-savvy
 - = Only emerging experience can throw light on whether such a business mix has had a favorable or unfavorable impact on the company's experience as compared with industry experience
- Financial projections: Sales and marketing expenses
 - + Primary marketing strategy is likely to be advertising therefore lesser expenses towards hiring direct sales staff and on-field sales force

- + This would be reduced recurring expenses under the new model in the form of salaries and performance related incentives
- Policyholder behaviour / moral hazard and adverse selection (Financial projections)
 - - There could be greater moral hazard as there is limited or no in-person verification of information furnished on proposal form questionnaires and the online quote inputs could be engineered to get the best quote
 - - potential concentration of risk arising from targeting a defined customer base could jeopardise risk selection
 - - This would need to be reflected in financial projections either by way of increased premiums or increased claim ratios
- Financial projections: Initial capital requirements
 - + Online sales would mean immediate access to cash on completion of sale (subject to any free look) which in turn:
 - Would reduce the need for greater initial capital needed for cash transactions like claim payments and expenses (subject to regulatory minimum capital required) vis-à-vis the regular model
 - Reduced requirements of capital infusions in future periods as may have been needed in the regular model
- Policyholder trust and data privacy (Business volumes)
 - = Online business is a faceless/impersonal business. Though the ecommerce client base is expected to be comfortable with online purchases, purchasing financial products and services online may not be viewed the same way by the same clients.
 - - Besides, since most sensitive personal information would have to be emailed / uploaded online, there could be data privacy concerns as compared with handing over these documents to a person
 - - This could potentially mean lower than expected conversion base of ecommerce clients into policyholders leading to lower business volumes
- Competitive pressures (Business volumes)
 - - Relying solely on online business may pose the threat of competitive poaching through aggressive marketing which is usually offline involving agents/intermediaries and direct sales force. This may adversely impact revenue.
- Opportunity cost of new strategy (Business volumes)
 - - The opportunity cost of not using brokers could be significant. If the loss of business going through traditional channels of brokers, OEMs in the market is significant, then business volumes will be negatively impacted and costs per policy might be excessively high to justify the strategy

[1 mark for each valid point with explanation]

[14]

vii) Addressing fraud risk [3 marks]:

Identifying sources of fraud (Risk Identification):

New Age Insurance Co is expected to be extensively operating digitally due to online sales only. This also subjects them to hacking, virus attacks, identity theft and other cyber risks.

Particularly in time of recession fraudulent activity is expected to be higher and New Age

Insurance Co needs to be more careful to safeguard itself through protection measures

Moral hazard and policyholder withholding / altering information or misrepresenting facts at time of purchase of policy or submitting a claim that are difficult to verify

Impact analysis (Risk Assessment)

Heat maps of frequency / severity to analyse extent of exposure to potential fraud and to ensure that the measures to be put in place are commensurate with the level of expected impact should an incident arise.

E.g., cyber risks like hacking or data theft / ransomware attacks can be low frequency but extremely high severity events which would necessitate significant investment in protection measures

Fraud policy (Risk policy and monitoring)

Establish a comprehensive fraud risk governance framework with defined procedures and protocols to identify and deal with fraud...

- Promote a zero-tolerance to fraud policy
- Set up dedicated fraud department
- Focus on training staff to identify fraudulent activity
- Clear feedback and reporting mechanism for monitoring and review

Fraud Management and minimisation measures (*Risk treatment*) [6 marks]

Fraud prevention

Strengthen underwriting standards:

- Tighten policy terms and conditions e.g. introduce deductibles or copayments
- Carefully draft policy wording with requirements to provide evidence of declarations made on proposal

Since all business is to be sold online, the company is expected to have a huge data repository of customer data captured at underwriting, policy details, exposure measures and claim history. The data can be utilized in following ways:

- Increase internal data analysis by tapping into this database for checks on claims and analysis for portfolio optimisation
- Consider sharing data with other insurers to identify duplicate coverage and multiple claimants

Educate policyholders e.g. wrongful or incomplete disclosure on proposals could invalidate claims and result in cancellation of cover by:

- Advertise cases where fraudsters were denied claims, future cover or even prosecuted
- Encourage the public to report insurance fraudsters through anonymous phone lines, with possible incentives to the whistle blowers
- Lobbying government (directly or through industry bodies) for stricter laws and

tougher penalties for insurance fraudsters

- Collaborate with police, media and other insurers to publicise specific example cases of fraud previously discovered
- Report fraud cases to the insurance regulator

Fraud detection

Stricter loss verification and claim settlement standards...

- Increase proportion of claims investigated prior to payment
- Conduct spot checks on claims of various sizes
- Refer large claims or suspect cases to a specialist fraud unit
- Cross-check policy coverage details with claim details e.g. cover dates against accident date, sum insured details, any specified exclusions in cover
- Check duplicate claims for the same loss event benefitting multiple policyholders
- Cross-check claim details with independent sources where possible
- Review claims for odd circumstances e.g. motor accidents during curfew, rain damage in dry weather days

Invest in technology to capture fraudulent behaviour...

- fraud waste abuse (FWA) software solutions to identify suspicious pattern of claims
- voice stress analysis systems for lie detection when reporting claims over telephonic calls
- data analytics to provide early indications on fraud prone segments of the portfolio
- looping analytics to underwriting and pricing so such segments can be charged a higher premium or be discontinued from further renewals
- use of machine learning and artificial intelligence for processing images and visual data to assess damage and quantify loss or validate claim amount

Fraud control

- Have some form of experience rating in place to reflect policyholder's claim experience in pricing his cover
- Require medical evidence from professionals
- Require original receipts or invoices to validate value of items stolen or damaged
- Consider settlement by replacement items instead of cash reimbursement of loss
- Consider restricting cashless cover to pre-approved hospital networks and provide reimbursement to non-network hospitals after detailed claim scrutiny
- Tighten claims handling procedures
- Check prior claim history of policyholder at inception and refer suspicious cases for further investigation e.g. accidents and claim records for vehicle and driver(s) and medical conditions / disease history for insured
- Use in-house loss adjusters / claim handlers for inspection where necessary

Fraud protection

- Need to have strong cyber security and safety mechanisms to prevent fraud from hackers altering policy and/or claims data records

- Need to safeguard personal and sensitive policyholder details e.g. medical records and financial bank or credit card details used for making premium payment online
- Consider purchasing cyber insurance cover for self

[9]

viii)

‘No reinsurance proposition’

Practicality:

In Indian context, obligatory quota share ceding to GIC would mean that a ‘no reinsurance’ plan is not possible. However, the insurer may decide to not purchase any reinsurance (proportional or non-proportional) over and above the obligatory ceding. Such a decision may require an explanation to be given to the regulator.

[1]

Impact:

To assess impact, need to look at the reduction in the net cost of reinsurance, so, account for the reduction in both reinsurance premium and recoveries. Also, need to assess the impact on profit with and without reinsurance arrangement.

[0.5]

Overall impact will also depend expected net loss ratios under various reinsurance arrangements relative to the gross loss ratios of the direct business.

[0.5]

Consider losing out on profit commissions if any from reinsurance arrangements. For instance, since it is a new company, under a sliding scale commission arrangement an initial provisional commission may be expected to be received from the reinsurer(s).

[0.5]

No reinsurance may increase volatility (depending on the line of business mix) and this may not be acceptable per the risk appetite of the company.

[0.5]

No reinsurance may lead to higher capital requirements and lower ratings.

[0.5]

Reinsurers can provide technical help with market knowledge, underwriting and claims administration which is especially useful for a new insurance company

[0.5]

[maximum 3 marks]

[3]

[52 Marks]
